

Bayesian Methods for Clinical Trials

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Practical 1: Bayesian sequential designs

You are asked to design a Phase II single-agent PoC clinical trial of a new anti-cancer agent. The team would like to evaluate a particular dose of the new agent in terms of the objective response rate (ORR) at Week 12 after the start of the treatment. The uninteresting response rate that would not suggest a promising activity is $p_0 = 0.30$ and an interesting treatment effect that would trigger the further development of the compound is $p_1 = 0.50$. The currently planned sample size is $N = 40$ patients.

You are asked to design this study using a Bayesian design and consider inclusion of the interim analysis/analyses during the course of the trial. In the first instance, you can assume a weakly informative prior of $Beta(1, 1)$ for the response probability.

- (a) Firstly, consider a design with no interim analysis. Apply posterior probability criterion to claim the efficacy at the end of the trial. Specifically, use

$$\mathbb{P}(p > p_0) > c_{\text{efficacy}}$$

where c_{efficacy} is a pre-specified threshold for the decision making.

Find the critical probability threshold for the posterior probability that will ensure the control of the type I error at 5%. What power is achieved for the proposed sample size and the found value of the probability threshold? Would you recommend to change the sample size?

- (b) Consider now an inclusion of the interim analysis after the half of the total number of patients has been recruited in the trial. At the time of the interim analysis, one can stop the trial earlier for efficacy only. Use the same probability threshold found in (a) for the interim and final analysis. What effect will it have on the type I error? Do you need to adjust the critical probability threshold. If yes, find the new value that control the type I error at the desirable level. What is the power of the study now?
- (c) Consider now an inclusion of the interim analysis after the half of the total number of patients has been recruited in the trial. At the time of the interim analysis, one can stop the trial earlier for futility only. The decision rule for the futility is based on the posterior probability. Specifically, use

$$\mathbb{P}(p < p_0) > c_{\text{futility}}$$

i.e. the posterior probability of being *below* the uninteresting value exceeds the pre-specified threshold.

Evaluate several design options with different futility bounds using thresholds $c_{futility} = 0.25; 0.50; 0.85$. Propose a design option that achieves a balance in the operating characteristics. Why have you chosen this design?

- (d) Consider now the futility rule based on the conditional power that uses the planned response rate (i.e. assumed under the alternative). Specifically, the conditional power should be

$$\mathbb{P}(X_{final} > X_{critical} | X_1 \text{ and } p = p_1) < c_{CP}$$

where $X_{critical}$ is the number of responses needed at the final analysis to reach the posterior probability success criterion; X_{final} is the number of responses at the final; and c_{CP} is the futility threshold.

Find a value of the futility bound that would match as closely as possible the operating characteristics of the design in point (c). Have you managed to find one or the operating characteristics could not be matched? Why?

- (e) Consider now the futility rule based on the predictive power (aka assurance). Specifically, use the criterion:

$$\mathbb{P}(X_{final} > X_{critical} | X_1 \text{ and}) < c_{PP}$$

where c_{PP} is the futility threshold.

Evaluate several design options using different futility bounds. Find a value of the futility bound that would match as closely as possible the operating characteristics of the design in point (c).

- (f) Assume now that there is some prior knowledge about the response rate. Specifically, the most likely value is 40% and we are 80% confident that the response rate is between 20% and 65%. The Beta prior matching these characteristics is as follows: $Beta(3.2, 7.5 - 3.2)$

For the new prior distribution, evaluate the decision rules proposed in (c). How does the new prior change the operating characteristics? Why? Does one need to make any changes to the design under the new prior? Why?

- (g) Assume now that, for financial reasons, you are asked to investigate the sample size not of $N = 40$ patients as before but of $N = 38$ patients. Under the design evaluated in point (b) [i.e. stopping earlier for efficacy only at the half of the planned total sample size], find the new operating characteristics. Is type I error controlled under the unadjusted values? Under the previously adjusted value? Why?

Do the same evaluation under the sample size of $N = 42$. What is the conclusion now?

- (g) Assume now that another inclusion criterion has been added in the protocol, and now the null hypothesis is that the uninteresting treatment effect is $p_0 = 0.15$. Is the critical value found in point (a) still suitable and controls the type I error at the desirable level? What would be the conclusion if the new interesting treatment effect would be $p_0 = 0.50$?